IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2025

MATHEMATICS FOR SIGNALS AND SYSTEMS SOLUTIONS

SOLUTIONS

1. (a) i. Given the size of the vectors, we realise that $\mathbf{A}$ has size 3×2 and a possible solution is

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix}.$$

ii. $\mathbf{A}$ has size 2×2 and a possible $\mathbf{A}$ is

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ a & b \end{bmatrix}.$$

We then need to impose

$$\begin{bmatrix} 3 & a \\ 1 & b \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \mathbf{0}.$$

This yields $3 + 3a = 0$ and $3b + 1 = 0$. Consequently,

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -1 & -1/3 \end{bmatrix}.$$

iii. Given the size of the vectors we realise that $\mathbf{A}$ must have size 3×2. However, the rank of $\mathbf{A}$ is 2 meaning the null space must be trivial and so requirements cannot be met.

(b) i. We denote with $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ the SVD of $\mathbf{A}$. We then have

$$\mathbf{C} = \mathbf{W}\mathbf{A}\mathbf{Q} = \mathbf{W}\mathbf{U}\mathbf{\Sigma}\mathbf{Q}\mathbf{V}^T$$

where clearly $\mathbf{WU}$ and $\mathbf{QV}^T$ are both unitary matrices by construction. So, the SVD of $\mathbf{C}$ is $\mathbf{C} = \hat{\mathbf{U}}\mathbf{\Sigma}\hat{\mathbf{V}}^T$ with $\hat{\mathbf{U}} = \mathbf{WU}$ and $\hat{\mathbf{V}}^T = \mathbf{QV}^T$. Since $\mathbf{\Sigma}$ is in common, this shows that $\mathbf{A}$ and $\mathbf{C}$ have the same singular values.

ii. We compute

$$\mathbf{A}^T\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Consequently the two singular values are $\sigma_1 = \sqrt{3}$ and $\sigma_2 = \sqrt{2}$. Since $\mathbf{A}^T\mathbf{A}$ is symmetric, its eigenvectors are orthogonal and are $v_2 = [1, 0]^T$ and $v_1 = [0, 1]^T$. It is then easy to find the three singular vectors $\mathbf{u}_1, \mathbf{u}_2$ and $\mathbf{u}_3$ as follows:

$$\mathbf{u_1} = \frac{1}{\sqrt{3}}\mathbf{A}\mathbf{v}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\mathbf{u_2} = \frac{1}{\sqrt{2}}\mathbf{A}\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{u_3} = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix},$$

where we found $\mathbf{u}_3$ by imposing orthonormality of the three vectors.

iii. We use (i) to conclude that the singular values of $\mathbf{C}$ are $\sigma_1 = \sqrt{3}$ and $\sigma_2 = \sqrt{2}$ and that singular vectors are the columns of :

$$\mathbf{WU} = \frac{1}{\sqrt{6}} \begin{bmatrix} 0 & \sqrt{6} & 0 \\ \sqrt{2} & 0 & -2 \\ 2 & 0 & \sqrt{2} \end{bmatrix}$$

and the rows of

$$\mathbf{QV}^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

(c) i.

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & 0 \\ -100 & 100 \end{bmatrix}.$$

So

$$\mathbf{x}_1 = \mathbf{A}^{-1}\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

ii. When the true value of $\mathbf{b}_1$ is corrupted by noise $\mathbf{e} = [0.1, 0.2]^T$, we have that

$$\mathbf{x}_2 = \mathbf{A}^{-1}\mathbf{b}_2 = \begin{bmatrix} 1.1 \\ 10 \end{bmatrix}.$$

So a small perturbation of $\mathbf{b}$ leads to a big change in $\mathbf{x}$. Specifically

$$\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|}{\|\mathbf{x_1}\|} = 10.0005 > \frac{\|\mathbf{b}_1 - \mathbf{b}_2\|}{\|\mathbf{b}_1\|} = 0.1581.$$

This is not surprising, since we know that

$$\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|}{\|\mathbf{x_1}\|} \leq \mathcal{K} \frac{\lambda_{max}\|\mathbf{b}_1 - \mathbf{b}_2\|}{\lambda_{min}\|\mathbf{b}_1\|},$$

where $\mathcal{K}$ is the condition number of $\mathbf{A}$ and in our case $\mathcal{K} = \sqrt{\lambda_{max}/\lambda_{min}} \approx 200$. Here λ_{max} and λ_{min} are the eigenvalues of $\mathbf{A}^T\mathbf{A}$.

iii. The regularized version of the problem

$$\min_x \left(\|\mathbf{Ax} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|^2 \right).$$

is equivalent to

$$\min_x \left(\|\tilde{\mathbf{A}}x - \tilde{\mathbf{b}}\|^2 \right).$$

with

$$\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{A} \\ \lambda \mathbf{I} \end{bmatrix}$$

and

$$\tilde{\mathbf{b}} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}.$$

The matrix $\tilde{\mathbf{A}}$ is tall and full column rank, so we are faced with a standard least-square problem that can be solved using the normal equation:

$$\tilde{\mathbf{A}}^T\tilde{\mathbf{A}}\mathbf{x}_\lambda = \tilde{\mathbf{A}}^T\tilde{\mathbf{b}}$$

which yields: $\mathbf{x}_\lambda = (\mathbf{A}^T\mathbf{A} + \lambda^2\mathbf{I})^{-1}\mathbf{A}^T\mathbf{b}$.

(d) We can solve this problem using a trial and error approach. Otherwise we can focus on the condition number of

$$\mathbf{B} = \mathbf{A}^T\mathbf{A} + \lambda^2\mathbf{I}$$

and we want the ratio between the largest and smallest eigenvalue of $\mathbf{B}$ to be smaller than 5 (please note that this is just a guideline). We note:

$$\mathbf{B} = \begin{bmatrix} 2+\lambda^2 & 0.01 \\ 0.01 & 0.001+\lambda^2 \end{bmatrix}.$$

This matrix is symmetric and close to being diagonal and so with good approximation we have that $\lambda_{max} \approx 2 + \lambda^2$ and $\lambda_{min} \approx 0.01 + \lambda^2$ So if we pick $\lambda = 1$ than $\lambda_{max}/\lambda_{min} \approx 3$ and so this choice meets our requirement. We can in fact immediately see that:

$$\frac{\|\mathbf{x}_1 - \mathbf{x}_\lambda\|}{\|\mathbf{x_1}\|} = 0.3334 \leq \frac{5|\mathbf{b}_1 - \mathbf{b}_2\|}{\|\mathbf{b}_1\|} = 0.79.$$

2. (a) i. FALSE. The fact that $n > m$ does not guarantee that the m vectors are independent. For example assume $n = 3$ and $m = 2$ we can have the vector in V be $\mathbf{v}_1 = [1, 1, 1]^T$ and $\mathbf{v}_2 = [2, 2, 2]^T$ which are clearly linearly dependent.

ii. TRUE. Vectors are linearly dependent if a combination of them (using some non-zero coefficients) gives the zero vector. This can obviously always be achieved if one of the vectors is the zero vector.

iii. TRUE since if two vectors in T were dependent then two vectors in V would also be dependent.

iv. TRUE since T is included in V and so the linearly dependent vectors of T also belong to V.

(b) i. We are trying to minimize $\|\mathbf{Ca} - \mathbf{y}\|^2$ with

$$\mathbf{C} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix},$$

$\mathbf{a} = [a_0, a_1]^T$ and $\mathbf{y} = [0.5, 1, 1.5, 2]$. Since $\mathbf{C}$ is full column rank, the system $\mathbf{Ca} = \mathbf{y}$ may have a solution and in that case the solution is unique. It is easy to see by inspection that in this case the solution exists and is given by $a_0 = 0$ and $a_1 = 0.5$. This solution is obviously also solving the least-square problem.

ii. The solution we found in part (i) does not satisfy the constraint $a_0 + a_1 t = 3.25$ for $t = 5$. So we need to solve the following constrained minimization: $\min_a \|\mathbf{Ca} - \mathbf{y}\|^2$ s.t. $\mathbf{Aa} = 3.25$ with $\mathbf{A} = [1\ \ 5]$. We use Lagrangian multipliers to solve this problem: minimize $L(\mathbf{a}, \lambda) = \|\mathbf{Ca} - \mathbf{y}\|^2 + \lambda(\mathbf{Aa} - 3.25)$. Taking the derivative over a_0, a_1 and λ and equating to zero yields:

$$2\mathbf{C^T Ca} - 2\mathbf{C^T y} + \mathbf{A}^T\lambda = \mathbf{0}$$

and

$$\mathbf{Aa} = 3.25.$$

Putting all together yields

$$\begin{bmatrix} 2\mathbf{C^T C} & \mathbf{A}^T \\ \mathbf{A} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{2C^T y} \\ 3.25 \end{bmatrix}.$$

In our case:

$$\begin{bmatrix} 2\mathbf{C^T C} & \mathbf{A}^T \\ \mathbf{A} & 0 \end{bmatrix} = \begin{bmatrix} 8 & 20 & 1 \\ 20 & 60 & 5 \\ 1 & 5 & 0 \end{bmatrix}$$

and

$$\begin{bmatrix} \mathbf{2C^T y} \\ 3.25 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \\ 3.25 \end{bmatrix}.$$

This leads to the following solution:

$$\begin{bmatrix} a_0 \\ a_1 \\ \lambda \end{bmatrix} = \frac{1}{60} \begin{bmatrix} 25 & -5 & -40 \\ -5 & 1 & 20 \\ -40 & 20 & -80 \end{bmatrix} \begin{bmatrix} 10 \\ 30 \\ 3.25 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.75 \\ -1 \end{bmatrix}.$$

It is easy to see that the new polynomial $0.75t - 0.5$ satisfies the constraint $0.75t - 0.5 = 3.25$ for $t = 5$.

(c) i. We are told that $\mathbf{Q}$ is symmetric, therefore $p_{2,1} = p_{1,2} = 0.125$ and $p_{3,1} = p_{1,3} = 0.25$. Moreover, we know that $\mathbf{Q}$ is a Markov matrix so its columns sum up to one. So we have that:

$$\mathbf{Q} = \begin{bmatrix} 0.625 & 0.125 & 0.25 \\ 0.125 & 1 - 0.125 - \alpha & \alpha \\ 0.25 & \alpha & 1 - 0.25 - \alpha \end{bmatrix}.$$

A Markov matrix has at least one eigenvalue $\lambda_1 = 1$ and we also know the a second eigenvalue $\lambda_2 = 0.5$. Given that $\det(\mathbf{Q}) = 0.125$, we have $\lambda_1\lambda_2\lambda_3 = \det(\mathbf{Q}) = 0.125$. Consequently, $0.5\lambda_3 = 0.125$ or $\lambda_3 = 0.25$. Finally, using the property that $\text{trace}(\mathbf{Q}) = \lambda_1 + \lambda_2 + \lambda_3$, we obtain $2.25 - 2\alpha = 1.75$ which yields $\alpha = 0.25$ and

$$\mathbf{Q} = \begin{bmatrix} 0.625 & 0.125 & 0.25 \\ 0.125 & 0.625 & 0.25 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}.$$

ii. The steady state is the eigenvector $\mathbf{u}_1$ with eigenvalue $\lambda_1 = 1$. So $\mathbf{Q}\mathbf{u}_1 = \mathbf{u}_1$ and it is easy to see that $\mathbf{u}_1 = [1/3, 1/3, 1/3]^T$.

iii. Each town will have a population of $30,000$.

3. (a) The solution is $\mathbf{x}_{MN} = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{b}$. We have:

$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}.$$

and

$$\mathbf{x}_{MN} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1/5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2/5 \\ 1/5 \end{bmatrix}.$$

(b) i. Let $\beta := \dfrac{2}{\boldsymbol{v}^T\boldsymbol{v}}$. Then

$$\boldsymbol{H}(\boldsymbol{v}) \;=\; \boldsymbol{I} - \beta\, \boldsymbol{v}\boldsymbol{v}^T.$$

Since $(\boldsymbol{v}\boldsymbol{v}^T)^T = \boldsymbol{v}\boldsymbol{v}^T$, we have

$$\boldsymbol{H}(\boldsymbol{v})^T = (\boldsymbol{I} - \beta\, \boldsymbol{v}\boldsymbol{v}^T)^T = \boldsymbol{I} - \beta\, \boldsymbol{v}\boldsymbol{v}^T = \boldsymbol{H}(\boldsymbol{v}).$$

So it suffices to show $\boldsymbol{H}(\boldsymbol{v})^2 = \boldsymbol{I}$. [1/3]

$$\boldsymbol{H}(\boldsymbol{v})^2 = (\boldsymbol{I} - \beta\, \boldsymbol{v}\boldsymbol{v}^T)\,(\boldsymbol{I} - \beta\, \boldsymbol{v}\boldsymbol{v}^T) = \boldsymbol{I} - 2\beta\, \boldsymbol{v}\boldsymbol{v}^T + \beta^2\,(\boldsymbol{v}\boldsymbol{v}^T)(\boldsymbol{v}\boldsymbol{v}^T).$$

For any vector $\boldsymbol{v}$,

$$(\boldsymbol{v}\boldsymbol{v}^T)(\boldsymbol{v}\boldsymbol{v}^T) = \boldsymbol{v}\,(\boldsymbol{v}^T\boldsymbol{v})\,\boldsymbol{v}^T = (\boldsymbol{v}^T\boldsymbol{v})\,\boldsymbol{v}\boldsymbol{v}^T.$$

Hence

$$\boldsymbol{H}(\boldsymbol{v})^2 = \boldsymbol{I} - 2\beta\, \boldsymbol{v}\boldsymbol{v}^T + \beta^2(\boldsymbol{v}^T\boldsymbol{v})\, \boldsymbol{v}\boldsymbol{v}^T = \boldsymbol{I} - \big(2\beta - \beta^2(\boldsymbol{v}^T\boldsymbol{v})\big)\, \boldsymbol{v}\boldsymbol{v}^T.$$

$$2\beta - \beta^2(\boldsymbol{v}^T\boldsymbol{v}) = \frac{4}{\boldsymbol{v}^T\boldsymbol{v}} - \frac{4}{(\boldsymbol{v}^T\boldsymbol{v})^2}\,(\boldsymbol{v}^T\boldsymbol{v}) = 0.$$

Therefore,

$$\boldsymbol{H}(\boldsymbol{v})^2 = \boldsymbol{I} \quad\Longrightarrow\quad \boldsymbol{H}(\boldsymbol{v})^T\boldsymbol{H}(\boldsymbol{v}) = \boldsymbol{I}.$$ [3/3]

ii. A. First compute the two inner products

$$\boldsymbol{v}^T\boldsymbol{a} = \big(\boldsymbol{a} + \alpha\, \boldsymbol{e}_1\big)^T\boldsymbol{a} = \boldsymbol{a}^T\boldsymbol{a} + \alpha\, a_1 = \|\boldsymbol{a}\|_2^2 + \alpha\, a_1,$$

and

$$\boldsymbol{v}^T\boldsymbol{v} = \big(\boldsymbol{a} + \alpha\, \boldsymbol{e}_1\big)^T\big(\boldsymbol{a} + \alpha\, \boldsymbol{e}_1\big) = \boldsymbol{a}^T\boldsymbol{a} + 2\alpha\, a_1 + \alpha^2 = \|\boldsymbol{a}\|_2^2 + 2\alpha\, a_1 + \alpha^2.$$

Because $\alpha^2 = s^2\|\boldsymbol{a}\|_2^2 = \|\boldsymbol{a}\|_2^2$, we have

$$2\big(\|\boldsymbol{a}\|_2^2 + \alpha\, a_1\big) = \|\boldsymbol{a}\|_2^2 + 2\alpha\, a_1 + \alpha^2 = \boldsymbol{v}^T\boldsymbol{v}.$$ [2/4]

Now apply $\boldsymbol{H}(\boldsymbol{v})$ to $\boldsymbol{a}$:

$$\boldsymbol{H}(\boldsymbol{v})\,\boldsymbol{a} = \boldsymbol{a} - 2\,\frac{\boldsymbol{v}\,\boldsymbol{v}^T\boldsymbol{a}}{\boldsymbol{v}^T\boldsymbol{v}} = \boldsymbol{a} - \frac{2\big(\|\boldsymbol{a}\|_2^2 + \alpha\, a_1\big)}{\boldsymbol{v}^T\boldsymbol{v}}\,\boldsymbol{v} = \boldsymbol{a} - \boldsymbol{v} = -\,\alpha\, \boldsymbol{e}_1 = -\,s\,\|\boldsymbol{a}\|_2\,\boldsymbol{e}_1.$$

This shows that all components $2, \ldots, m$ of $\boldsymbol{H}(\boldsymbol{v})\,\boldsymbol{a}$ are zero and the first component equals $-s\,\|\boldsymbol{a}\|_2$. [4/4]

(c) i. Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ and denote its first column by

$$\boldsymbol{a}_1 := \boldsymbol{A}\boldsymbol{e}_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} \in \mathbb{R}^m.$$

Our goal is to construct a Householder reflector $\boldsymbol{H}_1 \in \mathbb{R}^{m\times m}$ such that

$$\boldsymbol{H}_1\, \boldsymbol{a}_1 = \begin{bmatrix} \pm\|\boldsymbol{a}_1\|_2 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

We proceed in three steps.

Step 1: Let $s := \operatorname{sign}(a_{11})$, and set the scalar

$$\alpha := s\,\|\boldsymbol{a}_1\|_2.$$

Define the vector

$$\boldsymbol{v} := \boldsymbol{a}_1 + \alpha\,\boldsymbol{e}_1 \in \mathbb{R}^m.$$

[1/4]

Step 2: Define the Householder matrix

$$\boldsymbol{H}_1 := \boldsymbol{I} - 2\,\frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}} \quad \in \mathbb{R}^{m\times m}.$$

Then $\boldsymbol{H}_1$ is orthogonal ($\boldsymbol{H}_1^T\boldsymbol{H}_1 = \boldsymbol{I}$) and symmetric. A direct calculation as from the previous question gives

$$\boldsymbol{H}_1\boldsymbol{a}_1 = \boldsymbol{a}_1 - 2\,\frac{\boldsymbol{v}(\boldsymbol{v}^T\boldsymbol{a}_1)}{\boldsymbol{v}^T\boldsymbol{v}} = \boldsymbol{a}_1 - \frac{2\big(\|\boldsymbol{a}_1\|_2^2 + \alpha A_{11}\big)}{\|\boldsymbol{a}_1\|_2^2 + 2\alpha A_{11} + \alpha^2}\,\boldsymbol{v} = \boldsymbol{a}_1 - \boldsymbol{v} = -\alpha\,\boldsymbol{e}_1 = -\,s\,\|\boldsymbol{a}_1\|_2\,\boldsymbol{e}_1.$$

Thus the first column of $\boldsymbol{H}_1\boldsymbol{A}$ equals $(\pm\|\boldsymbol{a}_1\|_2, 0, \ldots, 0)^T$ (the sign is determined by s).

[3/4]

Step 3: Partition $\boldsymbol{A}$ by columns as

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{A}_{(:,\,2:n)} \end{bmatrix} \quad \text{with} \quad \boldsymbol{A}_{(:,\,2:n)} \in \mathbb{R}^{m\times(n-1)}.$$

Left-multiplying by $\boldsymbol{H}_1$ acts on all rows but preserves the zero pattern we created in column 1:

$$\boldsymbol{H}_1\boldsymbol{A} = \begin{bmatrix} \boldsymbol{H}_1\boldsymbol{a}_1 & \boldsymbol{H}_1\boldsymbol{A}_{(:,\,2:n)} \end{bmatrix} = \begin{bmatrix} \pm\|\boldsymbol{a}_1\|_2 & \boldsymbol{r}_{12}^T \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{bmatrix},$$

where

- $\boldsymbol{r}_{12}^T$ is the first row of $\boldsymbol{H}_1\boldsymbol{A}_{(:,\,2:n)}$ (a $1\times(n-1)$ row vector),

- $\boldsymbol{A}_{22}$ is the submatrix consisting of rows 2:m of $\boldsymbol{H}_1\boldsymbol{A}_{(:,\,2:n)}$, hence $\boldsymbol{A}_{22} \in \mathbb{R}^{(m-1)\times(n-1)}$.

By construction, all entries in rows 2:m of the *first* column are zero, i.e.,

$$\boldsymbol{H}_1\boldsymbol{A} = \begin{bmatrix} * & \boldsymbol{r}_{12}^T \\ \boldsymbol{0} & \boldsymbol{A}_{22} \end{bmatrix},$$

which is exactly the claimed form. [4/4]

ii. First column $\boldsymbol{a}_1 = [2,2,1]^T$ has norm $\|\boldsymbol{a}_1\| = 3$. Take $\boldsymbol{v}_1 = \boldsymbol{a}_1 + \|\boldsymbol{a}\|\boldsymbol{e}_1 = [5,2,1]^T$ and $\beta_1 = 2/(\boldsymbol{v}_1^T\boldsymbol{v}_1) = 1/15$. Then

$$\boldsymbol{H}_1 = \boldsymbol{I} - \beta_1\boldsymbol{v}_1\boldsymbol{v}_1^T = \begin{bmatrix} -\frac{2}{3} & -\frac{2}{3} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{11}{15} & -\frac{2}{15} \\ -\frac{1}{3} & -\frac{2}{15} & \frac{14}{15} \end{bmatrix}$$

[3/4]

then

$$\boldsymbol{A}^{(1)} = \boldsymbol{H}_1\boldsymbol{A} = \begin{bmatrix} -3 & -\frac{1}{3} & -4/3 \\ 0 & \frac{19}{15} & -14/15 \\ 0 & \frac{17}{15} & 23/15 \end{bmatrix}.$$

[4/4]